

Experiment 6

Aim: Write a program to simulate Deadlock management in distributed systems (Banker's)

Theory:

- Deadlock occurs in a distributed system when a group of processes waits indefinitely for resources held by other processes, forming a circular wait.
- Deadlock Conditions :-
 - 1) Mutual exclusion - A resource is held by one process at a time.
 - 2) Hold and wait - A process holds a resource and waits for more.
 - 3) No preemption - Resources cannot be taken away.
 - 4) Circular wait - A cycle of waiting processes exists.

Banker's Algorithm :-

- Banker's algorithm is a deadlock avoidance technique that prevents deadlock by allocating resources safely.
- Steps :-

1) Check request validity

→ A process requests resources. The system verifies if the request $\leq$ Need.

2) Simulate Allocation

- ⇒ The system temporarily allocates resources to the process.
- It runs a safety algorithm to check if the system remains in a safe state.

3) Grant or Deny Request

- ⇒ IF the system remains in a safe state, the request is approved.
- IF it leads to an unsafe state, the request is denied.

- Advantages :

- 1) Prevents deadlock by ensuring safe allocation.
- 2) Efficient for systems with known resource limits.
- 3) Works well for predictable workloads.

- Disadvantages :

- 1) Requires knowledge of maximum resource demand upfront.
- 2) High computational overhead in large systems.

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